

Titanic Exploratory Data Analysis (EDA) Report

Data Analyst Internship - Task 5

Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Titanic dataset to identify patterns, trends, relationships, and factors that influenced passenger survival.

Tools and Technologies Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook

Dataset Overview

The Titanic dataset contains demographic and travel-related information about passengers aboard the RMS Titanic. Important attributes include passenger age, gender, ticket fare, passenger class, and survival status.

Data Exploration Findings

Dataset Information: The dataset contains both numerical and categorical variables. A few columns contain missing values that require attention before advanced analysis.

Statistical Summary: Passenger ages vary across a wide range, while fare values show considerable variation. The difference between minimum and maximum fare values indicates the presence of outliers.

Missing Values: Missing values were identified primarily in the Age, Cabin, and Embarked columns. The Cabin column contained the largest number of missing records.

Visualization Insights

Survival Distribution

The number of passengers who did not survive was greater than the number who survived.

Age Distribution

Most passengers were between 20 and 40 years old, indicating a predominantly adult passenger population.

Fare Distribution (Boxplot)

Several outliers were identified, suggesting that some passengers paid significantly higher fares than others.

Survival by Gender

Female passengers had a considerably higher survival rate than male passengers.

Survival by Passenger Class

First-class passengers experienced the highest survival rates, while third-class passengers faced the highest mortality rates.

Correlation Heatmap

Relationships were observed between survival, passenger class, fare, and gender-related variables.

Pairplot Analysis

Pairwise relationships among age, fare, passenger class, and survival helped reveal trends and patterns in the data.

Key Findings

1. Female passengers were more likely to survive than male passengers.
2. Passenger class significantly influenced survival outcomes.
3. Most passengers belonged to the 20–40 age group.
4. Fare distribution contained notable outliers.
5. Missing values were concentrated in the Cabin column.
6. Correlation analysis revealed relationships between fare, class, and survival.

Conclusion

The Exploratory Data Analysis of the Titanic dataset revealed that gender and passenger class were the most influential factors affecting passenger survival. Through the use of statistical summaries and visualizations, meaningful patterns, trends, and relationships were identified, demonstrating the value of EDA in understanding real-world datasets.